

CS 170 Java Tools Summer 2016

This semester in CS 170 we'll be using the **DrJava** Integrated Development Environment. You're free to use other tools but this is what you'll use on your programming exams, so you'll need to be familiar with it. You also must use DrJava to check your homework assignments. The version of DrJava we'll be using has been *customized* for CS 170 to include built-in style checking and providing feedback and grading of your programming assignments. This short document will show you how to download and configure it on your own personal computer.

Step 1: Got Java?

The **Java Development Kit** (or **JDK**) is a free set of command-line tools available from Oracle. It is not an IDE or integrated development environment. If you download and install the JDK, it will also install the **Java Runtime Environment** or **JRE**, so you don't have to do both. Note that the JDK is a larger download than the JRE, however.

Here's how to install the JDK (get the latest version):

- Download **JDK8** (not the JRE) for your particular platform at:
<http://www.oracle.com/technetwork/java/javase/downloads/>

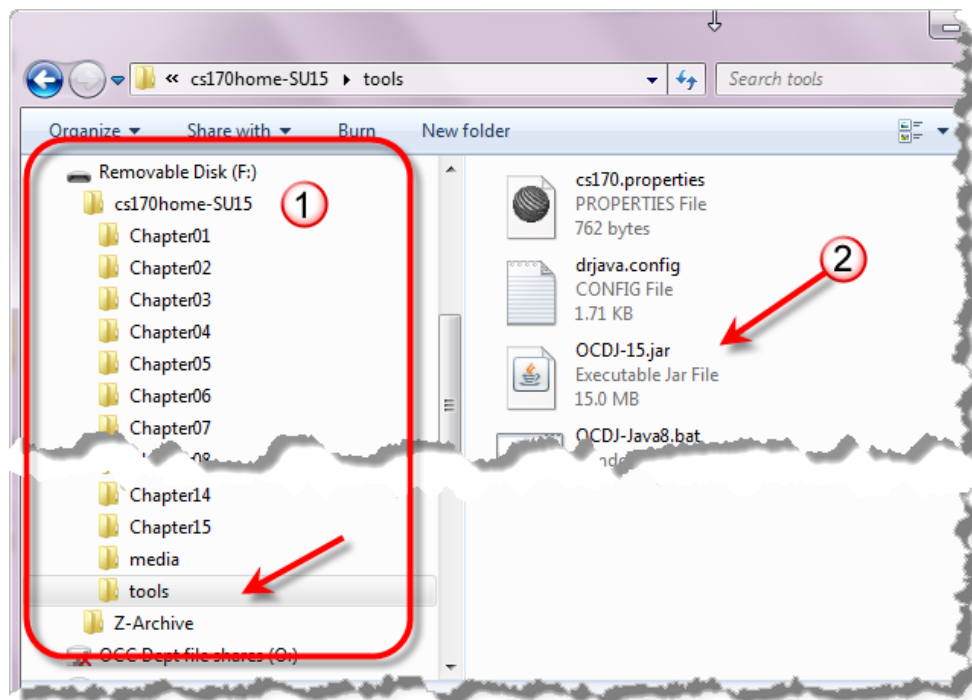


Then, follow the installation instructions from the Web page for your particular platform. While you can use Java 7, we are currently using Java 8 in the labs; there is no real reason to use Java 7. If you are using Windows-7-64-bit, then download the 64-bit version of the JDK.

Step 2: Call the Doctor

Start by downloading the Zip file **cs170home-SU16.zip** from Blackboard. You can put it on your desktop or in your documents folder. Once you've downloaded it, unzip it on your local machine. (Make sure that you actually *unzip* the file. Don't use the Windows feature that lets you treat the file as a compressed folder. If the zipped file looks like the picture here—with a zipper up the middle—then right-click it and choose **Extract All** from the context menu.)

Once you've unzipped **cs170home-SU16.zip** you'll find a folder named **cs170home**. This is the folder that contains **DrJava** and all of the starter files and test code that accompanies the online textbook we'll be using this semester. Go ahead and drag it wherever you want to keep it. In the picture shown below, you can see that I've dragged the folder to a USB thumb drive:



As you can see, the **cs170home** folder has a subfolder named **tools**, as well as folders for each chapter in the textbook. Open the **tools** folder and **double-click** our customized version of DrJava, **OCDJ-16.jar**. If all is well, you'll see the DrJava "splash screen" appear as DrJava loads for the first time.

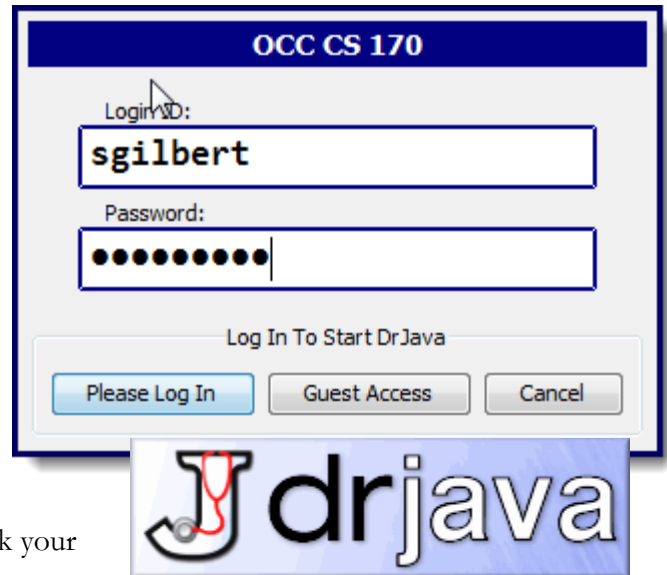


Logging In

When you first double-click **OCDJ-16.jar**, you should be taken to a **login dialog** that looks like this:

- Use your OCC **student email name**, (not the entire email address) as your login ID
- Use your **OCC student ID** (*including the capital C*) as your password.

If all goes well, you'll see the DrJava "splash screen" appear as DrJava loads for the first time. If you can't login for some reason, you can still use DrJava by clicking on **Guest Access**, but you won't be able to check your course progress unless you are able to log in.



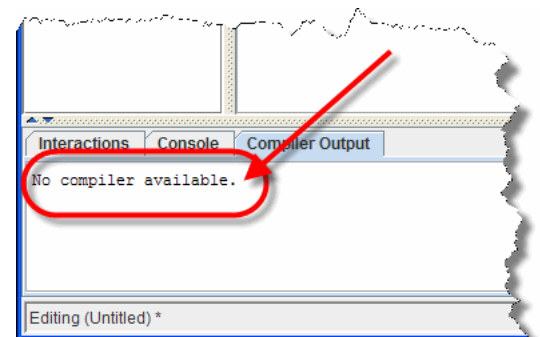
Troubleshooting Your Installation

DrJava doesn't start. If you have an "unarchiver" program like WinZip installed, it's possible for that program to "grab" the JAR (Java Archive) file extension, so that when you double-click the file it gets unzipped instead. (This may happen in different version of Linux as well.) If that occurs, open a command shell window, switch to the folder containing the Jar file, and type the command:
java -jar OCDJ-16.jar.

Moving the JAR: You *can* move the **OCDJ-16.jar** file, but it **must** be started from a folder named **tools** that also contains the **drjava.config** file from the original download.

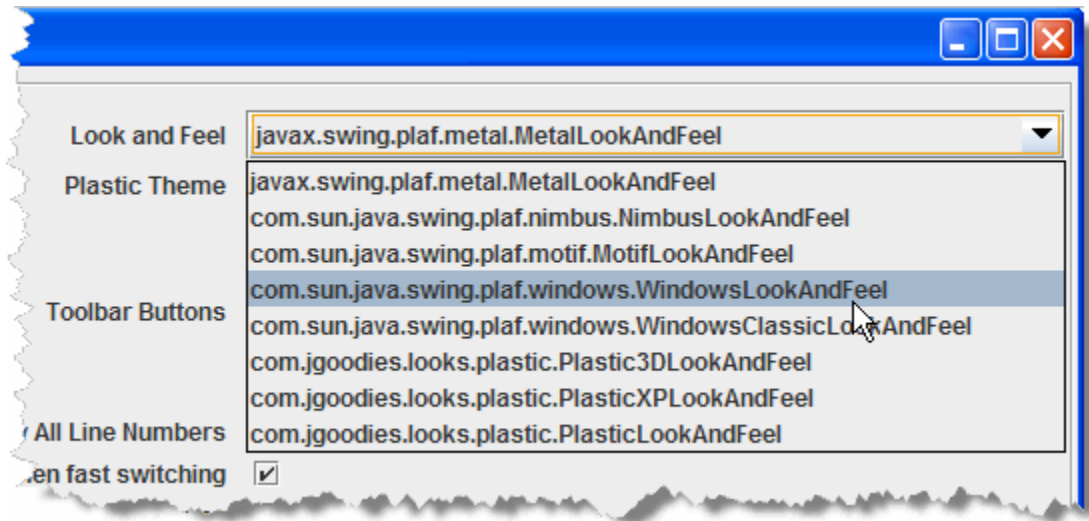
No compiler available. If you've tried the JRE only installation, when you start up DrJava on a machine without a JDK, you may see the message shown here when your program starts up. This should only occur on a machine that has an older version of Java installed.

The easiest fix is to make sure that you have a current version of Java installed by following the instructions in **Step 1: Got Java?**

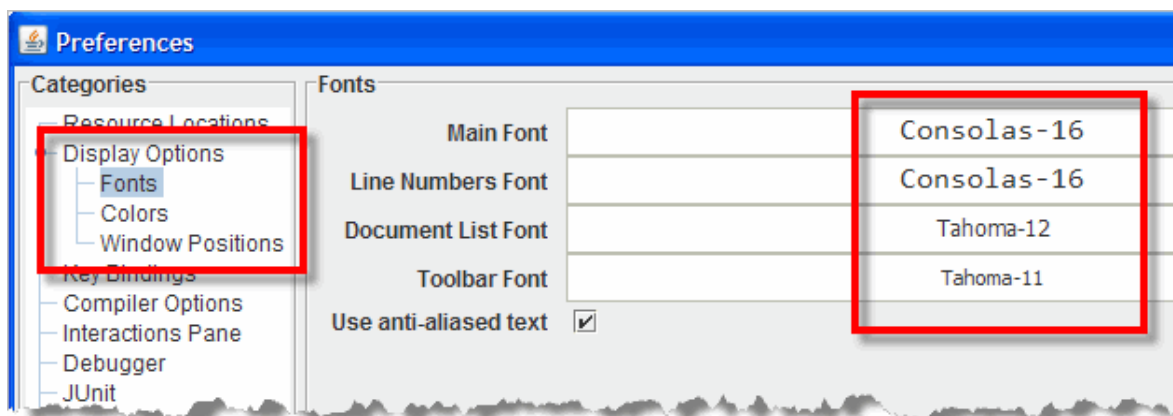


Step 3: Have It Your Way

I'll leave it to you to explore and to customize DrJava to your taste. Use the **Edit->Preferences** menu (on the Mac, you'll find it at **OCDrJava->Preferences** from the Mac menu at the top of the screen), make your changes. I've modified the CS 170 custom version of DrJava so that it uses the native look-and-feel of the Window system. You can switch to a different look and feel if you like:



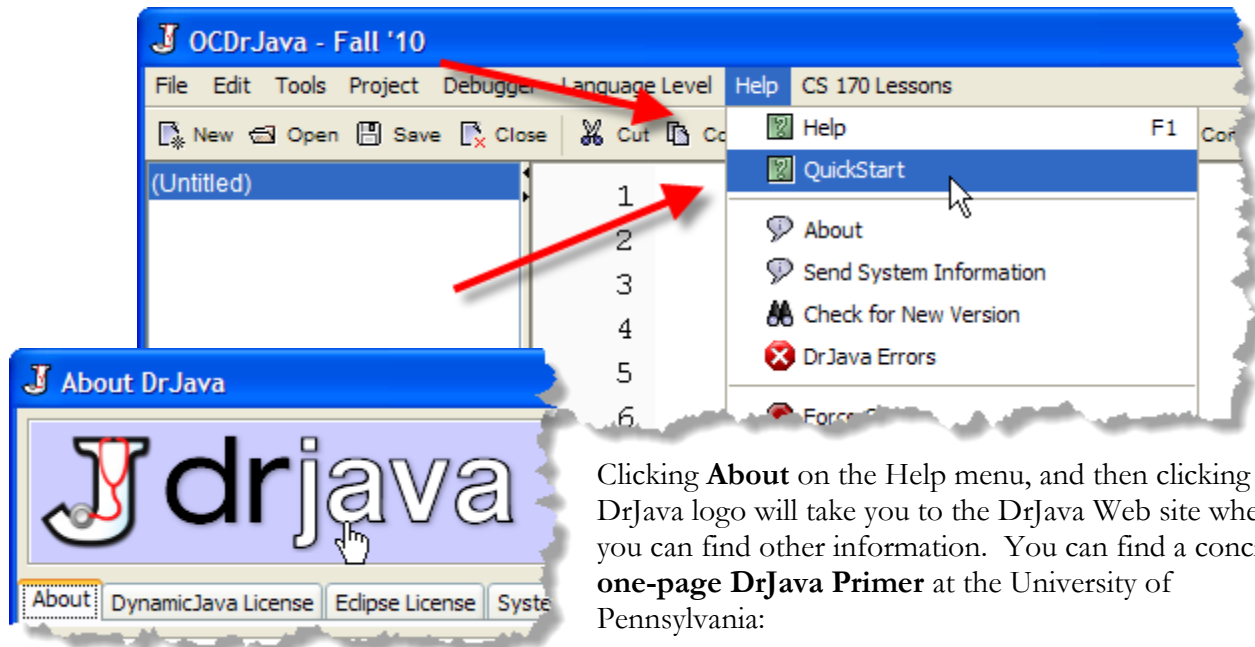
I also find that **the default fonts are really too small for me to see**. These are the settings I use on Windows. Unfortunately, when I take my USB drive over to my Mac, I then have to change the fonts to something else, like Monaco. If you leave the fonts at **Monospace**, then they'll work correctly, if unattractively, on either system:



Feel free to explore and change the other options like Colors, Window Positions and Key Bindings (hot-keys), to make DrJava work the way that you want it to work. You can't really hurt the program. If something goes horribly wrong, just replace the **.config** file with the original.

Step 4: Where to Go Now?

Browse the **DrJava Quick Start** from the **Help** menu to get a quick overview of using DrJava. Use **F1** or **Help->Help** to read the DrJava reference manual.



Clicking **About** on the Help menu, and then clicking the DrJava logo will take you to the DrJava Web site where you can find other information. You can find a concise, **one-page DrJava Primer** at the University of Pennsylvania:

<http://www.seas.upenn.edu/~cis1xx/resources/drjava/primer/>